

Compact gates / fewer sleeves and narrower paper

Suggested trial: **1000 mm opening, 12-inch paper**. 16 structural fittings + 16 sleeves = 32 prints. Compared with the current large gate: half the printed pieces, 19.0% less PETG and 21.3% less printer time. The 800 mm version needs the same prints but saves one PVC stick.

Opening	Paper	Outer mm	Sleeves / total	PETG kg	Print time	PVC sticks	Subtotal*
1000	24 in	2219.2	32 / 48	2.104	84h 20m	6	\$55.86
1000	18 in	1914.4	24 / 40	1.993	79h 18m	5	\$49.03
1000	12 in	1609.6	16 / 32	1.883	74h 21m	4	\$42.19
800	24 in	2019.2	32 / 48	2.104	84h 20m	5	\$50.52
800	18 in	1714.4	24 / 40	1.993	79h 18m	4	\$43.69
800	12 in	1409.6	16 / 32	1.883	74h 21m	3	\$36.85

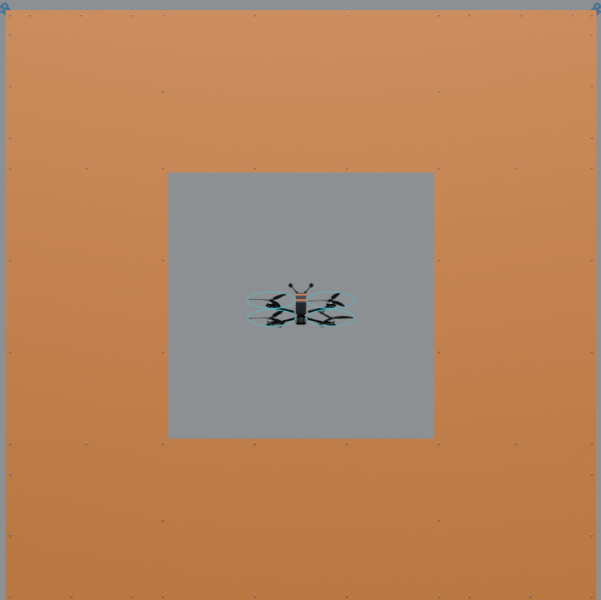
*PETG consumed + both magnets per clamp + whole PVC sticks, at the owner's purchase prices. Paper, guys and ground supports are excluded. Slicer estimates use exact batch quantities, including the two new partial sleeve plates.

The 350 mm maximum edge-magnet gap is an untested layout target. The 800 mm gate could use fewer long-rail sleeves only by accepting wider gaps (418 mm with one per rail). Test paper retention before trying that reduction. All six variants retain the accepted 33.5 mm fitting bores.

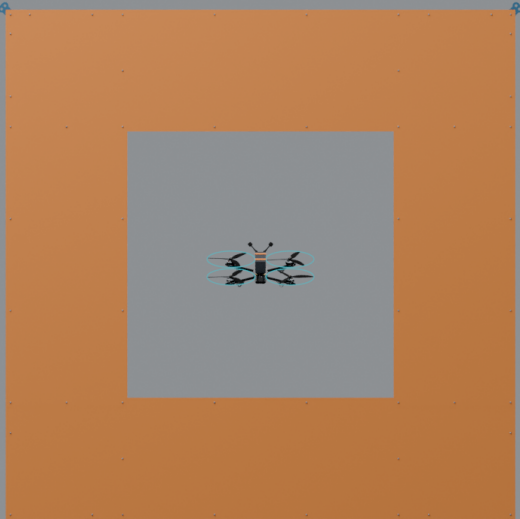
The accompanying README includes full print queues, pipe cuts, paper cuts and sleeve positions. Original fitting geometry is unchanged. Dendy / Printables 1515387 drone reference: CC BY-NC 4.0; battery assumed. Ground supports and guys are omitted from renders.

Smaller openings. Narrower paper. Fewer sleeves.

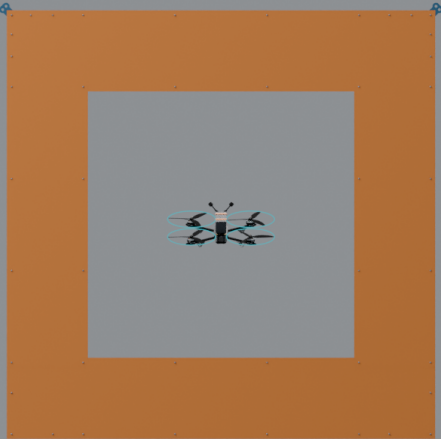
All six gates and drones use the same scale. 1000 mm openings above; 800 mm below. Bands: 24 / 18 / 12 inches.



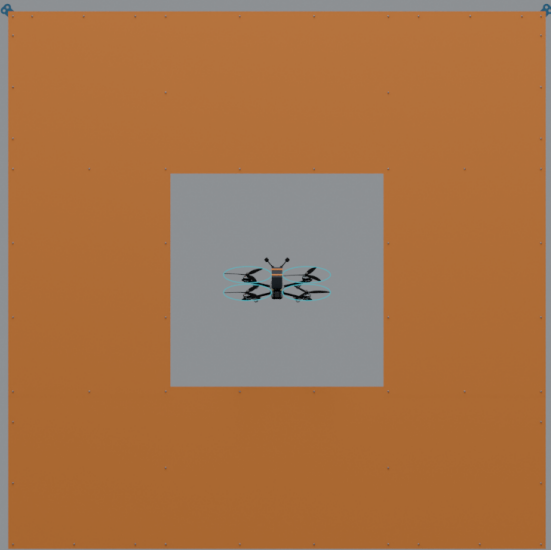
1000 mm / 24-inch paper
2219.2 mm outside / 32 sleeves



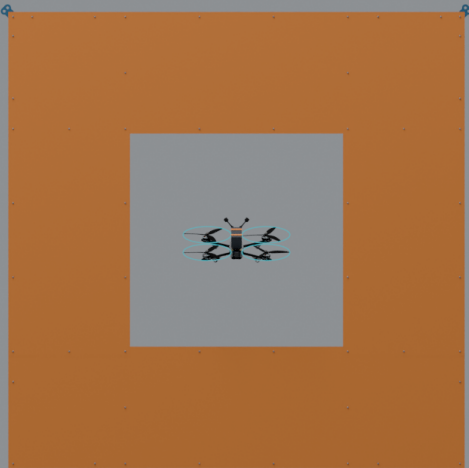
1000 mm / 18-inch paper
1914.4 mm outside / 24 sleeves



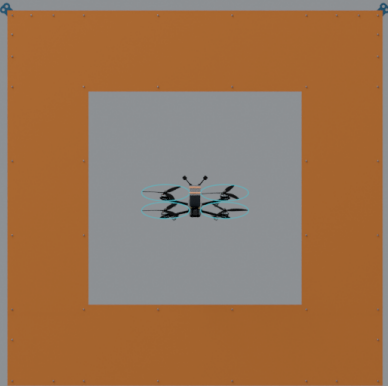
1000 mm / 12-inch paper
1609.6 mm outside / 16 sleeves



800 mm / 24-inch paper
2019.2 mm outside / 32 sleeves



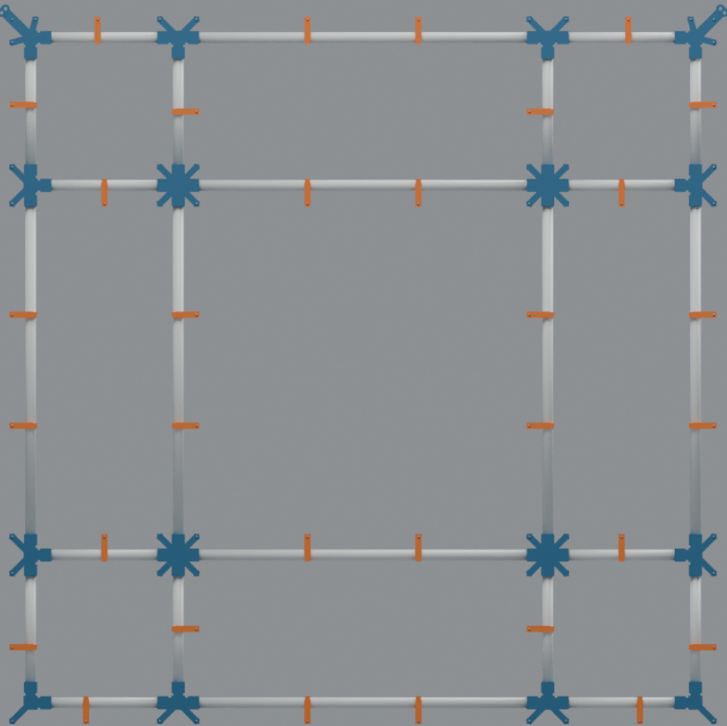
800 mm / 18-inch paper
1714.4 mm outside / 24 sleeves



800 mm / 12-inch paper
1409.6 mm outside / 16 sleeves

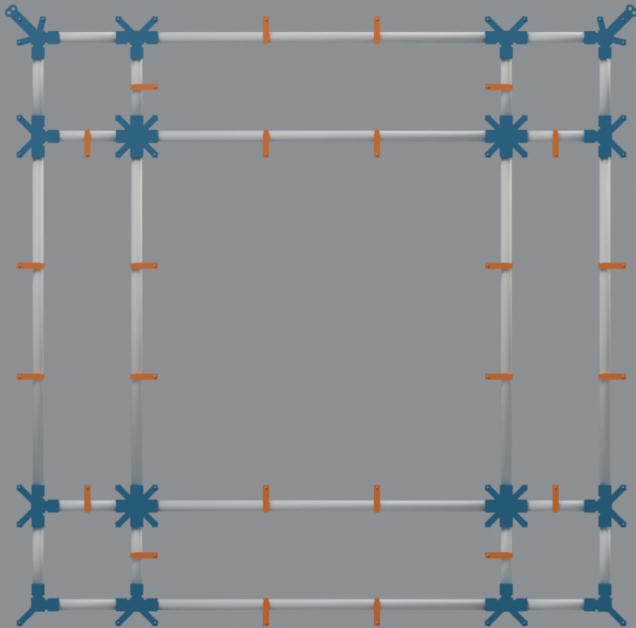
1000 mm opening / frame with paper removed

Blue: 16 unchanged structural fittings. Orange: reduced sleeve count. White: 1-inch PVC, accepted 33.5 mm sockets.



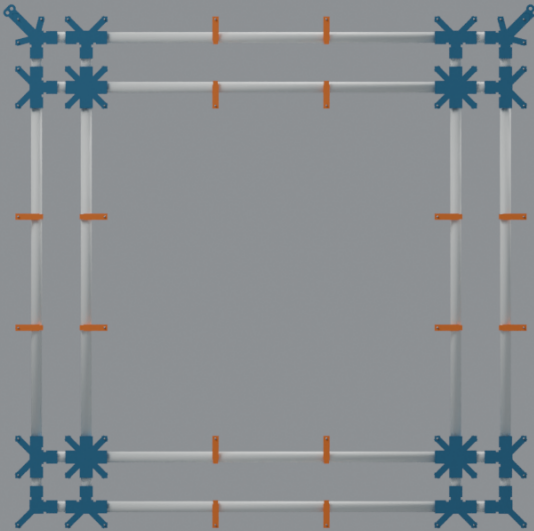
24-inch paper / 48 printed parts

32 orange sleeves / 6 PVC sticks



18-inch paper / 40 printed parts

24 orange sleeves / 5 PVC sticks



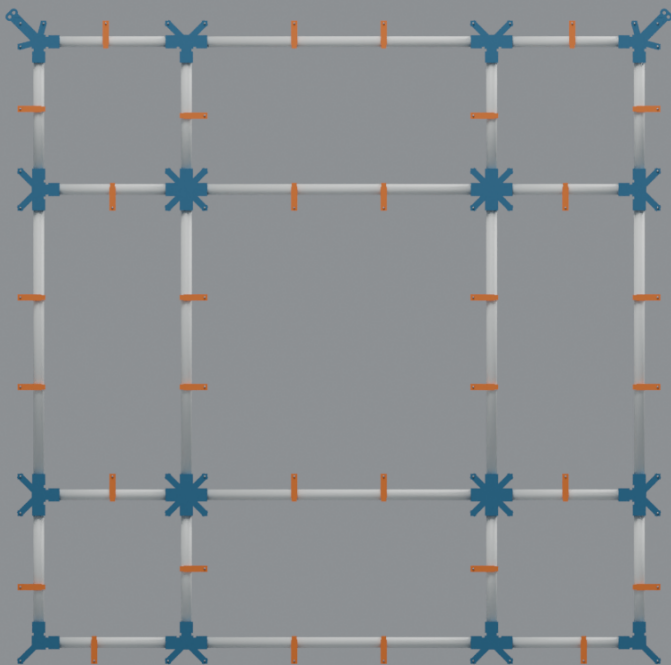
12-inch paper / 32 printed parts

16 orange sleeves / 4 PVC sticks

350 mm maximum magnet-gap trial. No internal cords, diagonals or additional fittings. Footings and external guys omitted.

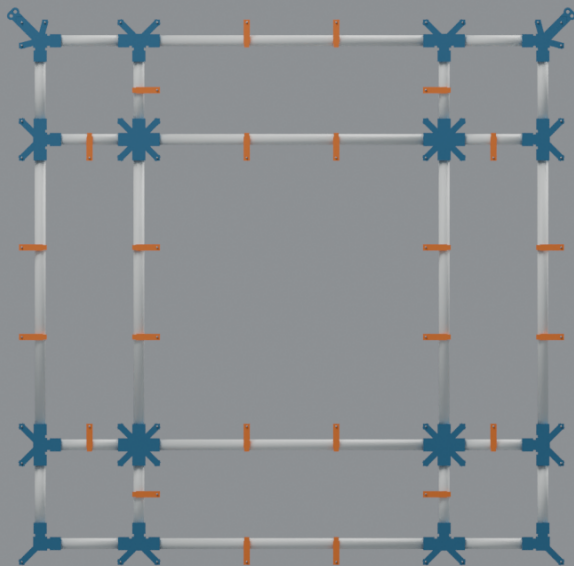
800 mm opening / frame with paper removed

Blue: 16 unchanged structural fittings. Orange: reduced sleeve count. White: 1-inch PVC, accepted 33.5 mm sockets.



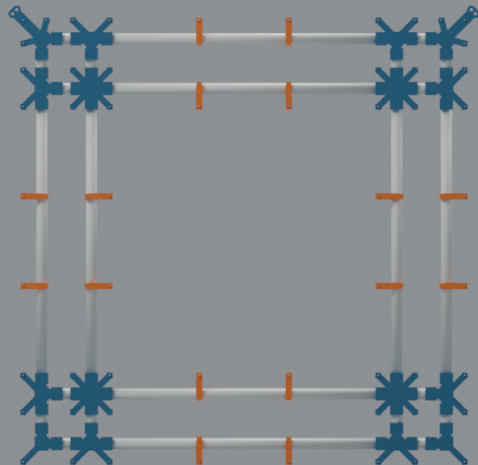
24-inch paper / 48 printed parts

32 orange sleeves / 5 PVC sticks



18-inch paper / 40 printed parts

24 orange sleeves / 4 PVC sticks



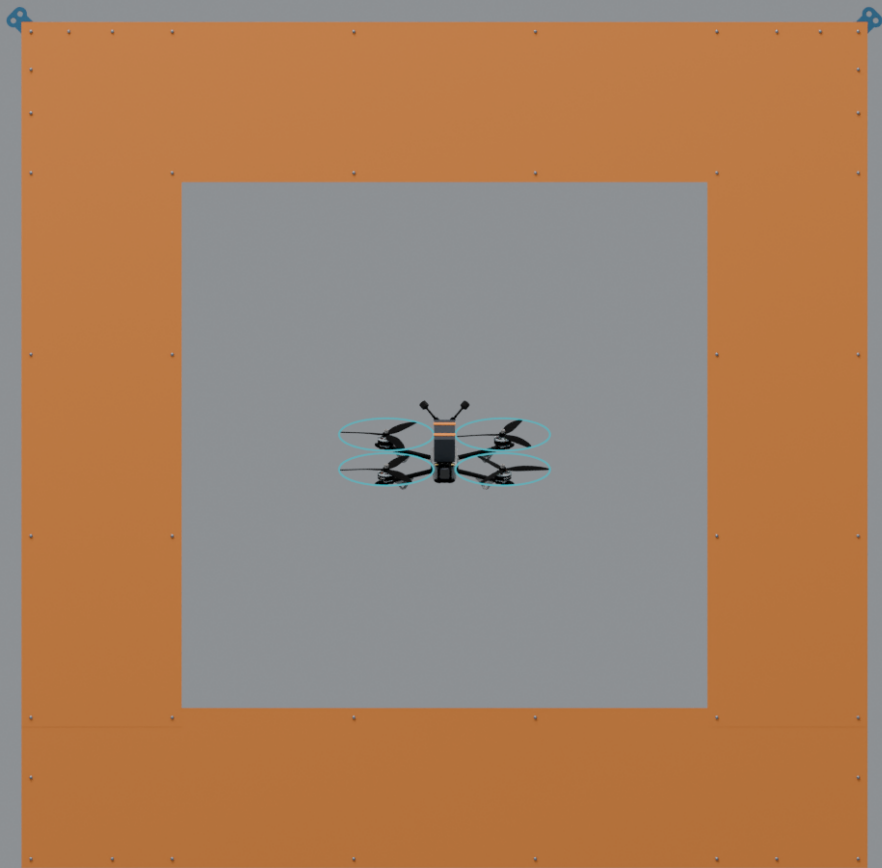
12-inch paper / 32 printed parts

16 orange sleeves / 3 PVC sticks

350 mm maximum magnet-gap trial. No internal cords, diagonals or additional fittings. Footings and external guys omitted.

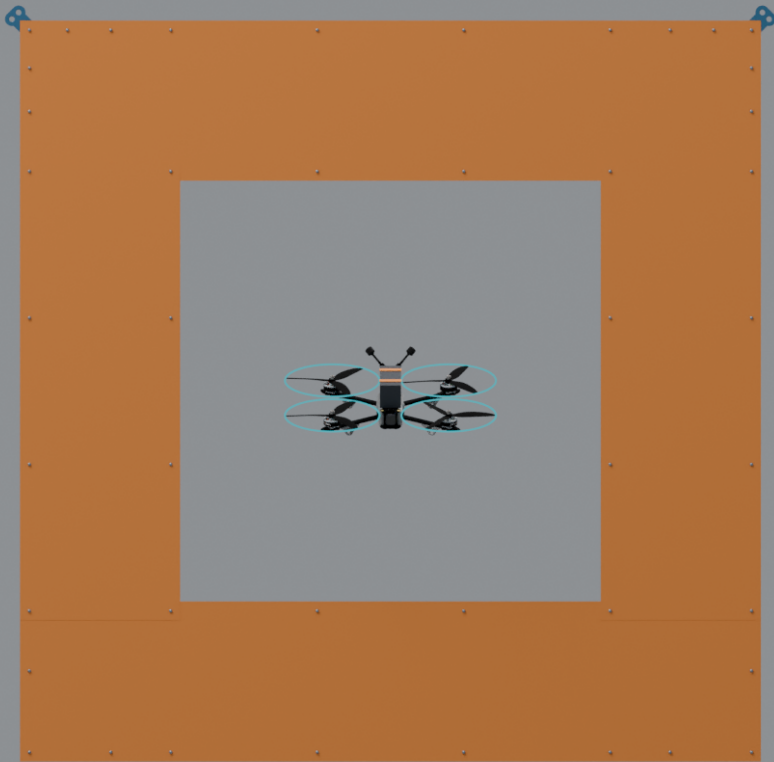
12-inch paper / the leanest compatible pair

Same drone scale. 1000 mm opening: 299 mm centered side gap. 800 mm opening: 199 mm side gap.



1000 mm opening / 1609.6 mm outside

16 sleeves + 16 fittings / 4 PVC sticks

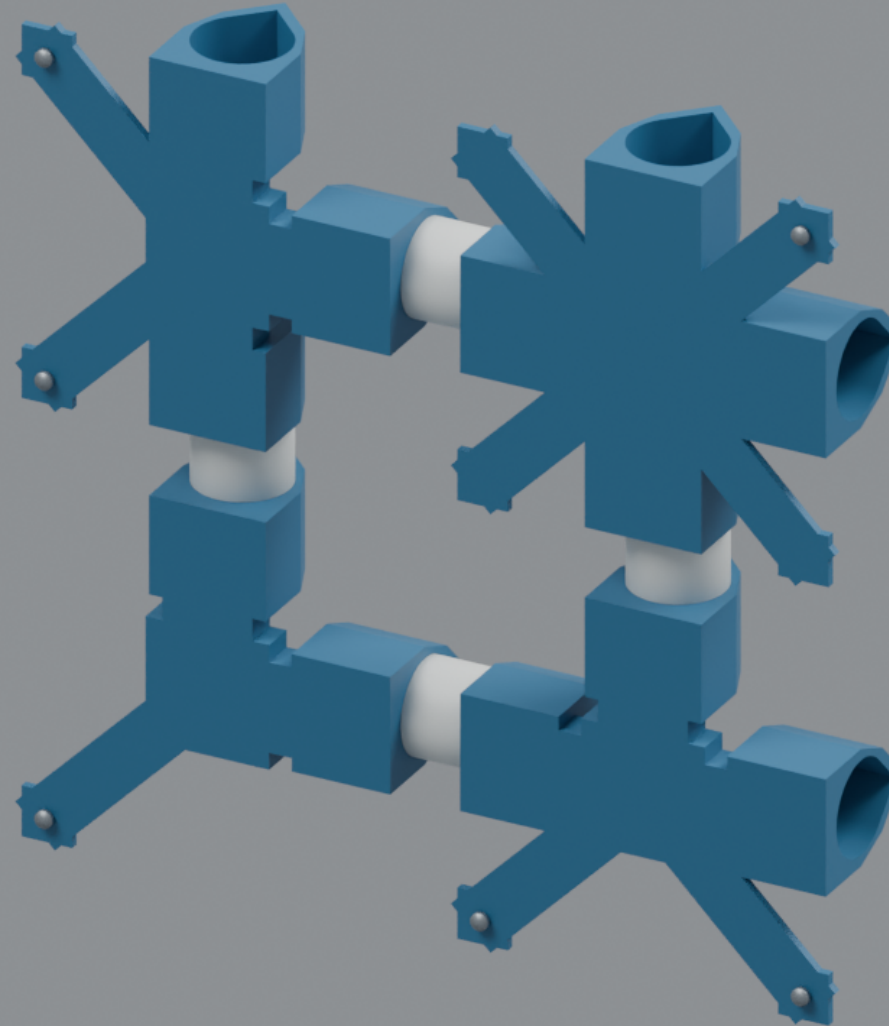


800 mm opening / 1409.6 mm outside

16 sleeves + 16 fittings / 3 PVC sticks

12-inch corner / no short-pipe sleeves

Four existing fittings are joined by 84.8 mm pipe cuts. 30 mm insertion per end leaves 24.8 mm of exposed pipe.



No new part geometry. Fitting-mounted magnets cover short edges and paper overlap seams; long rails still use sleeves.